COMPUTER ENGINEERING
QUEZON CITY

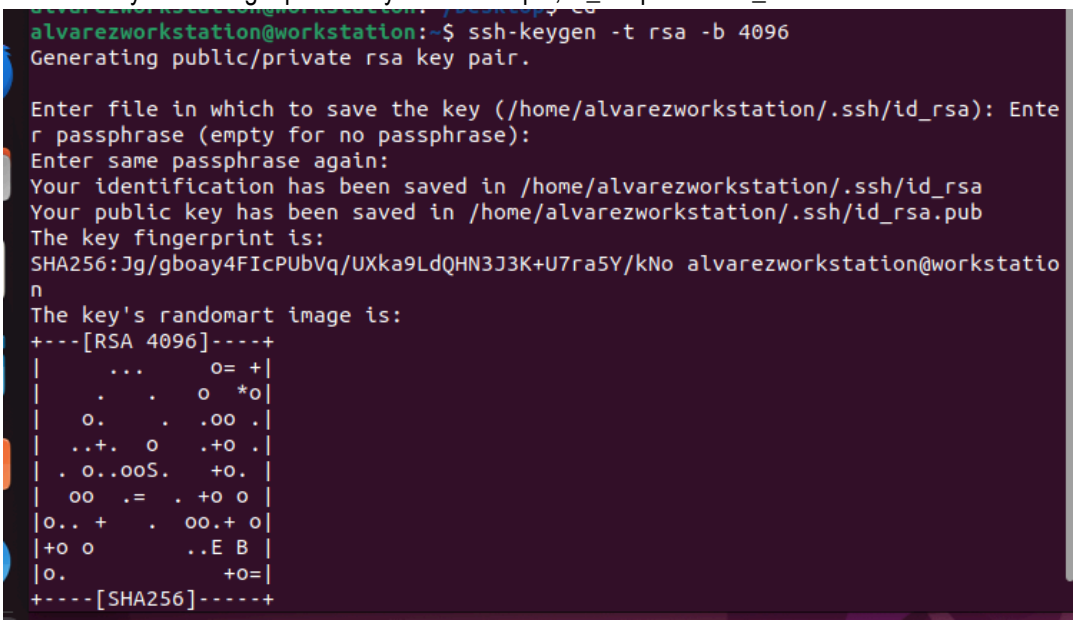
COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	07/30/26	**DATE SUBMITTED:**	07/30/26
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ACTIVITY NO. 2.1 **SSH Key-Based Authentication and GIT Setup**

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication

1. The simplest way to generate a key pair is to run ssh-keygen without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user authentication are usually stored in the users .ssh directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case id_rsa when using the default RSA algorithm. It could also be, for example, id_dsa or id_ecdsa.
2. Issue the command ssh-keygen -t rsa -b 4096. The algorithm is selected using the -t option and key size using the -b option.
3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that cannot be used even if someone obtains the private key file. The passphrase should be cryptographically strong.
4. Verify that you have created the key by issuing the command ls -la .ssh. The command should show the .ssh directory containing a pair of keys. For example, id_rsa.pub and id_rsa



Task 2: Copying the Public Key to the remote servers

1. To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh-copy-id tool.
2. Issue the command similar to this: ssh-copy-id -i ~/.ssh/id_rsa user@host
   Server 1:

```
alvarezworkstation@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa alvarezworkstatio
n@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/alvarezwork
station/.ssh/id_rsa.pub"
The authenticity of host 'server1 (192.168.56.120)' can't be established.
ED25519 key fingerprint is SHA256:k3OVu2x/fg+U2IY3lau+9ZaGtH7syLGSgj2eGBtdQqA.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
alvarezworkstation@server1's password:
```

Server 2:

```
alvarezworkstation@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa alvarezworkstatio
n@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/alvarezwork
station/.ssh/id_rsa.pub"
The authenticity of host 'server2 (192.168.56.121)' can't be established.
ED25519 key fingerprint is SHA256:gpnvZ4AQxgwSPr2+FY6OObdVCc3zERcDmOu0tCHcUoQ.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:7: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
alvarezworkstation@server2's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'alvarezworkstation@server2'"
```

3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.
4. On the local machine, verify that you can SSH with Server 1 and Server 2. What did you notice? Did the connection ask for a password? If not, why?
    - The ssh public key was already sent to the servers that's why it did not ask for the password.

Task 3: Set up the Git Repository

1. On the local machine, verify the version of your git using the command which git. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: sudo apt install git

```
alvarezworkstation@workstation:~$ sudo apt install git
[sudo] password for alvarezworkstation:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 4,154 kB of archives.
After this operation, 21.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 liberror-perl all 0.1
7029-1 [26.5 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 git-man all 1
:2.34.1-1ubuntu1.17 [954 kB]
21% [2 git-man 693 kB/954 kB 73%]                              90.8 kB/s 37s
```

2. After the installation, issue the command which git again. The directory of git is usually installed in this location: user/bin/git.

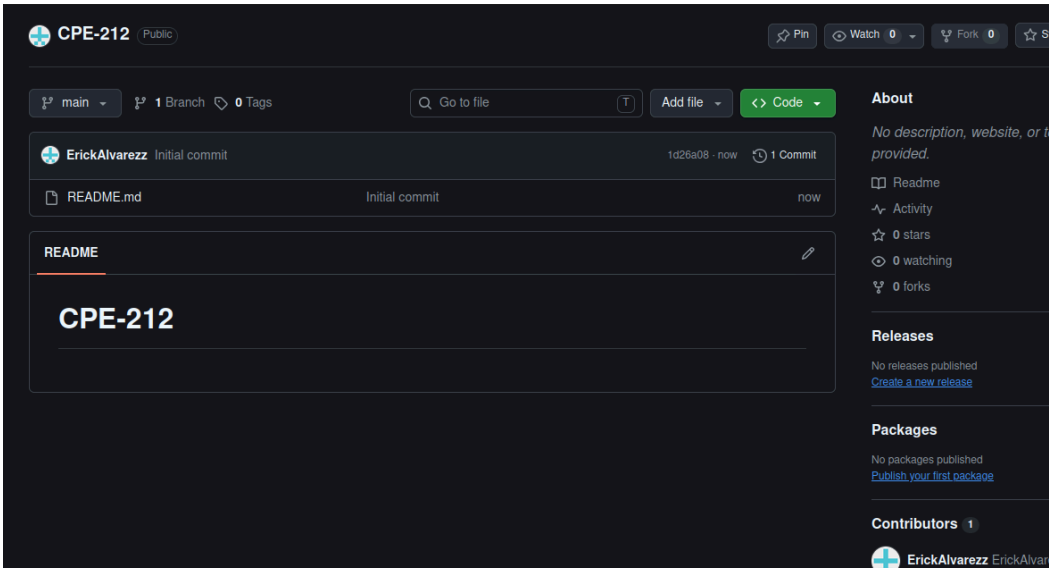
```
alvarezworkstation@workstation:~$ which git
/usr/bin/git
alvarezworkstation@workstation:~$
```

3. The version of git installed in your device is the latest. Try issuing the command git --version to know the version installed.

```
alvarezworkstation@workstation:~$ git --version
git version 2.34.1
alvarezworkstation@workstation:~$
```

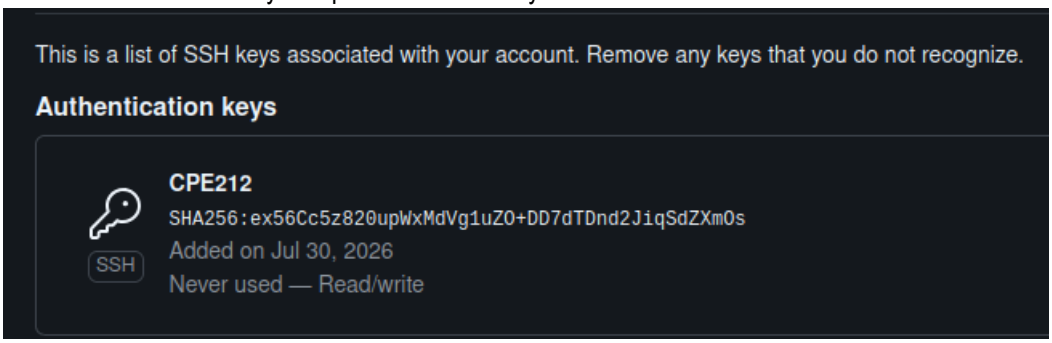
4.

5. Using the browser in the local machine, go to www.github.com.
6. Sign up in case you don't have an account yet. Otherwise, login to your GitHub account.
    a. Create a new repository and name it as CPE232_yourname. Check Add a README file and click Create repository.

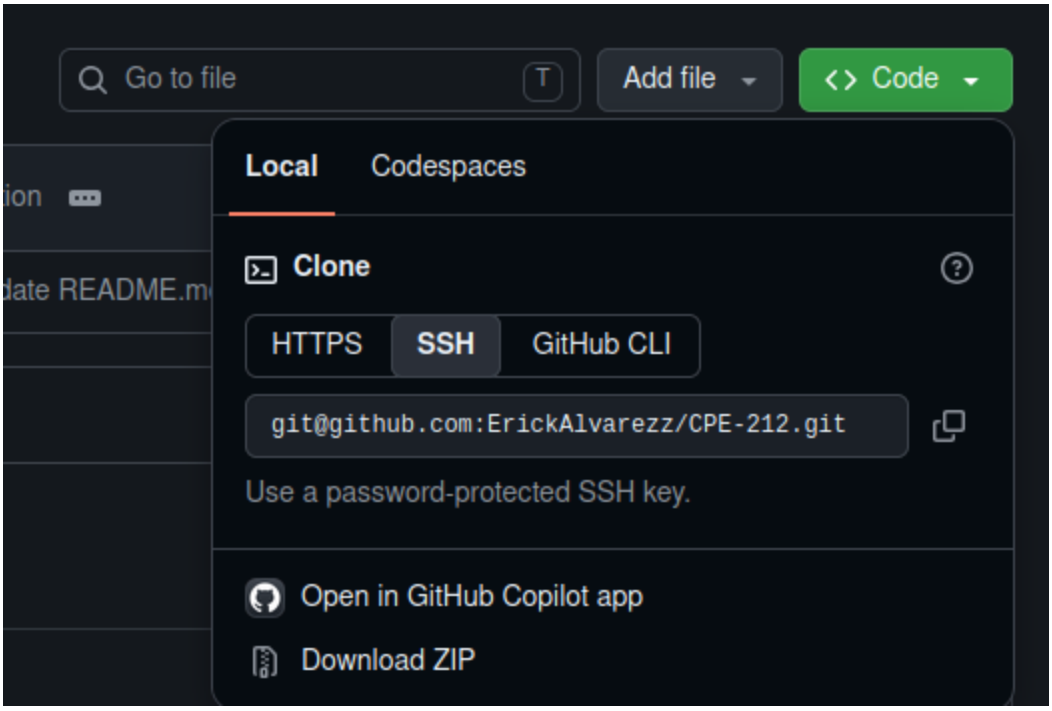


b. Create a new SSH key on GitHub. Go your profile's setting and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create a new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.

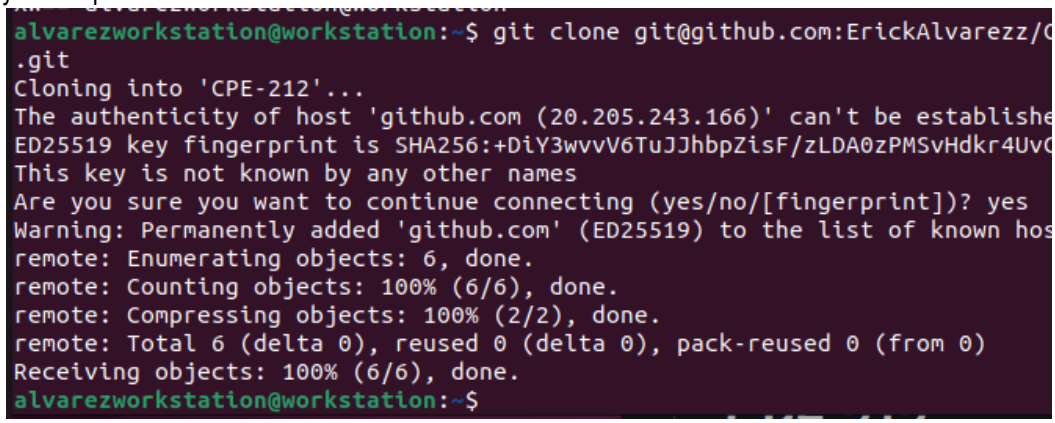
c. On the local machine's terminal, issue the command cat .ssh/id_rsa.pub and copy the public key. Paste it on the GitHub key and press Add SSH key.



d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the Code drop down menu. Select SSH and copy the link



e. Issue the command git clone followed by the copied link. For example, git clone git@github.com:jvtaylar-cpe/CPE232_yourname.git. When prompted to continue connecting, type yes and press enter.



f. To verify that you have cloned the GitHub repository, issue the command ls. Observe that you have the CPE232_yourname in the list of your directories. Use CD command to go to that directory and LS command to see the file README.md.

```
alvarezworkstation@workstation:~/CPE-212$ ls
README.md
alvarezworkstation@workstation:~/CPE-212$ cat README.cd
cat: README.cd: No such file or directory
alvarezworkstation@workstation:~/CPE-212$ cat README.md
# CPE-212
This is the README.md of CPE 212
alvarezworkstation@workstation:~/CPE-212$
```

g. Use the following commands to personalize your git.
- git config --global user.name “Your Name”
- git config --global user.email yourname@email.com
- Verify that you have personalized the config file using the command cat ~/.gitconfig

```
This is the README.md of CPE 212
alvarezworkstation@workstation:~/CPE-212$ git config --global user.name "Er
"
alvarezworkstation@workstation:~/CPE-212$ git config --global user.email qe
rez02@gmail.com
alvarezworkstation@workstation:~/CPE-212$ ls
README.md
alvarezworkstation@workstation:~/CPE-212$ cat ~/.gitconfig
[user]
        name = Erick A
        email = qejalvarez02@gmail.com
alvarezworkstation@workstation:~/CPE-212$
```

h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.

```
  GNU nano 6.2                        README.md *
# CPE-212
This is the README.md of CPE 212
ako si erick tagapagmana ng computer na to

File Name to Write: README.md
^G Help          M-D DOS Format    M-A Append        M-B Backup File
^C Cancel        M-M Mac Format    M-P Prepend       ^T Browse
```

i. Use the git status command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven’t, and which files aren’t

being tracked by Git. Status output does not show any information regarding the committed project history. What is the result of issuing this command?

```
alvarezworkstation@workstation:~/CPE-212$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
alvarezworkstation@workstation:~/CPE-212$
```

j. Use the command git add README.md to add the file into the staging area.

k. Use the git commit -m “your message” to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next commit.

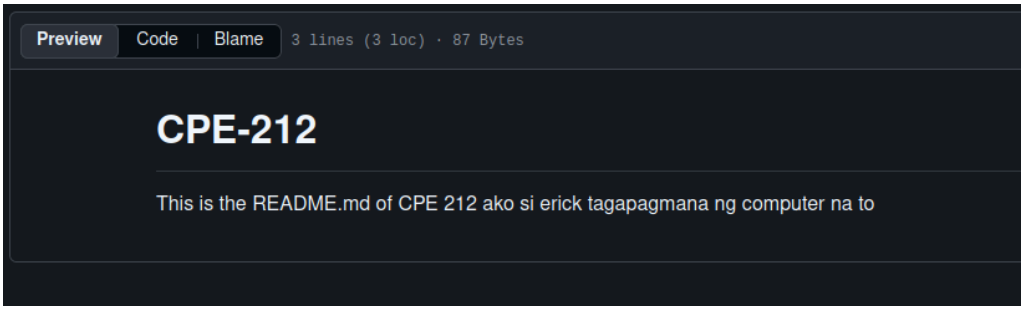
```
alvarezworkstation@workstation:~/CPE-212$ git add README.md
alvarezworkstation@workstation:~/CPE-212$ git commit -m "Hi, I'm anger"
[main 8da46c4] Hi, I'm anger
 1 file changed, 1 insertion(+)
alvarezworkstation@workstation:~/CPE-212$
```

l. Use the command git push <remote><branch> to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository. As an example, you may issue git push origin main.

```
alvarezworkstation@workstation:~/CPE-212$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 330 bytes | 330.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:ErickAlvarezz/CPE-212.git
   863e77b..8da46c4  main -> main
alvarezworkstation@workstation:~/CPE-212$
```

m. On the GitHub repository, verify that the changes have been made to README.md by refreshing

n. the page. Describe the README.md file. You can notice the how long was the last commit. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote.



Analysis: After editing it in my terminal, we can see that readme.md added some new sentence, “ako si erick tagapagmana ng computer na to”.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. How will you describe the ssh-program? What does it do?
   - SSH is an application that provides the user with the ability to log into remote devices. It secures the connection through encryption.
2. How do you know that you already installed the public key to the remote servers?
   - The fact that SSH was installed was confirmed since I could log in without a password.
3. What sort of things have we so far done to the remote servers using ansible commands?
   - Ansible was used to log into the remote servers, execute commands, and control the server from one computer.
4. How important is the inventory file?
   - The inventory file is important in our activity since it defines the remote servers to be managed by Ansible.

3. CONCLUSION

Through the activity of using SSH key based authentication and Git, I was able to gain knowledge on how to connect and manage remote server in a secure way. Through Ansible, I got the chance to know how to use it to control many servers from one computer. In general, the activity gave me insight about Linux.

4. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

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